PROJECT SYNOPSIS

1. Project Title

A Framework for Monitoring, Detecting, and Responding to Data Drift in Machine Learning Systems

2. Student Details

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3. Introduction / Background of the Study

Machine Learning models deployed in real-world environments often face changing data distributions over time, known as data drift. This leads to degradation in model performance and reliability. Traditional evaluation assumes static data and does not account for such shifts. This project proposes a framework that not only detects data drift using statistical techniques but also responds through a retraining mechanism. A dashboard interface is incorporated to monitor system behavior, making the solution closer to real-world ML system requirements.

4. Problem Statement

Machine learning models degrade when exposed to evolving data distributions. There is a need for systems that can detect drift, quantify its impact, and trigger corrective actions such as retraining.

5. Objectives of the Project

- Design a drift monitoring framework
- Implement KS Test and PSI
- Simulate drift scenarios
- Analyze performance degradation
- Implement retraining trigger
- Develop monitoring dashboard

6. Literature Review

Data drift and concept drift are widely studied challenges in machine learning deployment. Statistical techniques like KS Test and PSI are commonly used. Research in MLOps emphasizes continuous monitoring and retraining. However, integration of detection, response, and visualization in a single framework remains limited.

7. Scope of the Project

Focus on supervised ML models with structured data. Includes drift detection, retraining trigger, and dashboard visualization. Real-time distributed systems are out of scope.

8. Proposed Methodology

1. Requirement Analysis
2. Data Collection & Preprocessing
3. Model Development
4. Drift Simulation
5. Drift Detection (KS, PSI)
6. Retraining Trigger Mechanism
7. Dashboard Development
8. Evaluation & Analysis

9. Tools / Technologies to be Used

Python, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Plotly

10. Expected Outcomes

A complete framework for drift detection, automated retraining trigger, performance evaluation, and interactive dashboard.

11. References

1. Gama, J. et al. (2014). A Survey on Concept Drift Adaptation.
2. Lipton, Z. et al. (2018). Detecting and Correcting for Label Shift.
3. Sculley, D. et al. (2015). Hidden Technical Debt in ML Systems.
4. Baier, L. et al. (2020). Data Drift Detection in ML Models.
5. Webb, G. et al. (2016). Characterizing Concept Drift.

12. Approval Section

Role	Name	Signature	Date
Project Guide	Dr. Omkar Barve		